

## Aejandro De La Torre

LIS 440

wk14: More Graphing with Pandas/PyPlot

12/9/25

### Code

```
# Import the necessary packages
import pandas as pd
import os
from pathlib import Path
import matplotlib.pyplot as plt # Correct and consistent use

# Print version of pandas
print("Alejandro De La Torre's code....")
print(pd.__version__)

# -----
# File & Path Setup
# -----

# Set root directory (two levels up from this script)
root = Path(__file__).parent.parent.resolve()

# Show current working directory
print("Current Working Directory:", Path.cwd())

# Define key directories
data = root / 'data'
data_raw = data / 'raw'
data_processed = data / 'processed'
output = root / 'output' / 'figures'
output.mkdir(parents=True, exist_ok=True) # Make sure output folder exists

# Fetch the data
icecreamDF = pd.read_csv(data_processed / 'icecreamadditions.csv')
print("\nIce Cream Data: ")
print(icecreamDF)

# =====
# GRAPH 1: All Flavors vs. All Containers
# =====
print("\nGRAPH 1: All Flavors vs. All Containers...")
icecreamFlavorRowsDF = icecreamDF.set_index("Flavor")
print("icecreamFlavorRowsDF:")
print(icecreamFlavorRowsDF)

icecreamFlavorRowsDF.plot(kind="bar")
plt.title("All Flavors and All Containers")
plt.ylabel("Scoops")
plt.tight_layout()
plt.savefig(output / "graph1_all_flavors_all_containers_flavors_x.png")
plt.show()
```

```

# =====
# GRAPH 2: All Containers vs. All Flavors
# =====
print("\nGRAPH 2: All Containers vs. All Flavors...")
icecreamContainerRowsDF = icecreamFlavorRowsDF.transpose()
print("icecreamContainerRowsDF:")
print(icecreamContainerRowsDF)

icecreamContainerRowsDF.plot(kind="bar")
plt.title("All Containers and All Flavors")
plt.ylabel("Scoops")
plt.tight_layout()
plt.savefig(output / "graph2_all_containers_all_flavors_containers_x.png")
plt.show()

# =====
# GRAPH 3: Missouri Flavor Pie Chart
# =====
print("\nGRAPH 3: Missouri Pie Chart...")
selectedFlavor = "Missouri"
selectedFlavorSeries = icecreamFlavorRowsDF.loc[selectedFlavor]
print("selectedFlavorSeries:")
print(selectedFlavorSeries)

selectedFlavorSeries.plot(kind="pie", label="", autopct='%1.1f%%')
plt.title("Your State Favorite: Missouri")
plt.tight_layout()
plt.savefig(output / "graph3_missouri_pie_chart.png")
plt.show()

# =====
# GRAPH 4: Missouri Bar Chart by Container
# =====
print("\nGRAPH 4: Missouri Bar Chart by Container...")
selectedFlavorSeries.plot(kind="bar")
plt.title("Your State Favorite: Missouri - All Containers")
plt.ylabel("Scoops")
plt.tight_layout()
plt.savefig(output / "graph4_missouri_bar_chart.png")
plt.show()

# =====
# GRAPH 5: Selected Flavors (Strawberry, Cookie Dough, Missouri) vs Selected Containers
# (SugarCone, CakeCone)
# =====
print("\nGRAPH 5: Strawberry, Cookie Dough, Missouri vs. SugarCone and CakeCone
(Flavors on x-axis)...")
selectedFlavorsList = ["Strawberry", "Cookie Dough", "Missouri"]
selectedContainersList = ["SugarCone", "CakeCone"]

selectedFlavorsDF = icecreamFlavorRowsDF.loc[selectedFlavorsList]
selectedFlavorsAndContainersDF = selectedFlavorsDF[selectedContainersList]
print("selectedFlavorsAndContainersDF:")
print(selectedFlavorsAndContainersDF)

```

```

selectedFlavorsAndContainersDF.plot(kind="bar")
plt.title("Selected Flavors and Containers")
plt.ylabel("Scoops")
plt.tight_layout()
plt.savefig(output / "graph5_selected_flavors_containers_flavors_x.png")
plt.show()

# =====
# GRAPH 6: Selected Containers (SugarCone, CakeCone) vs Selected Flavors (Strawberry,
# Cookie Dough, Missouri)
# =====
print("\nGRAPH 6: SugarCone and CakeCone vs. Strawberry, Cookie Dough, Missouri
(Containers on x-axis)...")
selectedContainersDF = icecreamContainerRowsDF.loc[selectedContainersList]
selectedContainersAndFlavorsDF = selectedContainersDF[selectedFlavorsList]
print("selectedContainersAndFlavorsDF:")
print(selectedContainersAndFlavorsDF)

selectedContainersAndFlavorsDF.plot(kind="bar")
plt.title("Selected Containers and Flavors")
plt.ylabel("Scoops")
plt.tight_layout()
plt.savefig(output / "graph6_selected_containers_flavors_containers_x.png")
plt.show()

```

## Shell

```

>>> %Run pyplot_matplotlib_viz.py
Alejandro De La Torre's code....
2.3.3
Current Working Directory: /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S
440/Lectures/Week 14/ploplot_graphing/code

```

Ice Cream Data:

|   | Flavor       | SugarCone | Cup | WaffleCone | CakeCone |
|---|--------------|-----------|-----|------------|----------|
| 0 | Chocolate    | 44        | 34  | 26         | 49       |
| 1 | Strawberry   | 22        | 41  | 22         | 43       |
| 2 | Vanilla      | 33        | 31  | 44         | 96       |
| 3 | Blue Moon    | 11        | 29  | 22         | 100      |
| 4 | Cookie Dough | 91        | 48  | 39         | 95       |
| 5 | Missouri     | 82        | 51  | 72         | 96       |

GRAPH 1: All Flavors vs. All Containers...

icecreamFlavorRowsDF:

|              |  | SugarCone | Cup | WaffleCone | CakeCone |
|--------------|--|-----------|-----|------------|----------|
| Flavor       |  |           |     |            |          |
| Chocolate    |  | 44        | 34  | 26         | 49       |
| Strawberry   |  | 22        | 41  | 22         | 43       |
| Vanilla      |  | 33        | 31  | 44         | 96       |
| Blue Moon    |  | 11        | 29  | 22         | 100      |
| Cookie Dough |  | 91        | 48  | 39         | 95       |
| Missouri     |  | 82        | 51  | 72         | 96       |

GRAPH 2: All Containers vs. All Flavors...

icecreamContainerRowsDF:

| Flavor     | Chocolate | Strawberry | Vanilla | Blue Moon | Cookie Dough | Missouri |
|------------|-----------|------------|---------|-----------|--------------|----------|
| SugarCone  | 44        | 22         | 33      | 11        | 91           | 82       |
| Cup        | 34        | 41         | 31      | 29        | 48           | 51       |
| WaffleCone | 26        | 22         | 44      | 22        | 39           | 72       |
| CakeCone   | 49        | 43         | 96      | 100       | 95           | 96       |

GRAPH 3: Missouri Pie Chart...

selectedFlavorSeries:

SugarCone 82

Cup 51

WaffleCone 72

CakeCone 96

Name: Missouri, dtype: int64

GRAPH 4: Missouri Bar Chart by Container...

GRAPH 5: Strawberry, Cookie Dough, Missouri vs. SugarCone and CakeCone (Flavors on x-axis)...

selectedFlavorsAndContainersDF:

|              | SugarCone | CakeCone |
|--------------|-----------|----------|
| Flavor       |           |          |
| Strawberry   | 22        | 43       |
| Cookie Dough | 91        | 95       |
| Missouri     | 82        | 96       |

GRAPH 6: SugarCone and CakeCone vs. Strawberry, Cookie Dough, Missouri (Containers on x-axis)...

selectedContainersAndFlavorsDF:

| Flavor    | Strawberry | Cookie Dough | Missouri |
|-----------|------------|--------------|----------|
| SugarCone | 22         | 91           | 82       |
| CakeCone  | 43         | 95           | 96       |

>>>

Process ended with exit code 1.

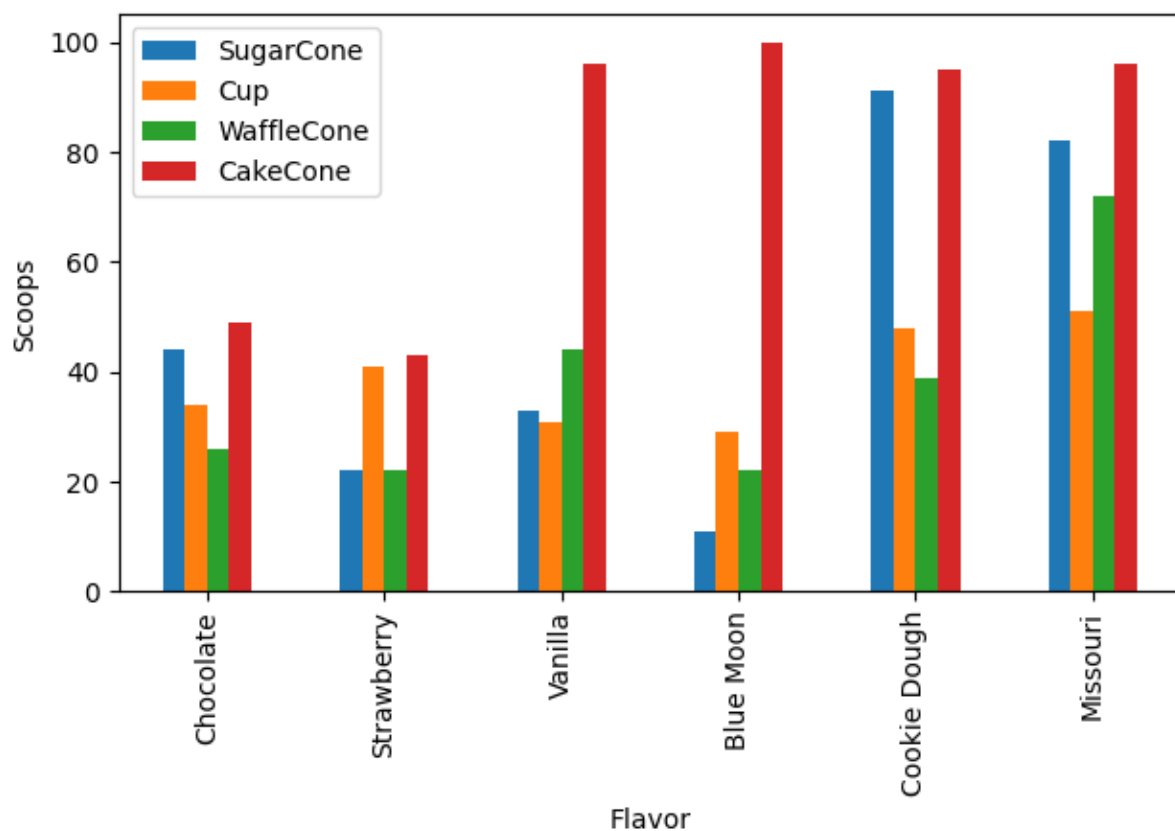
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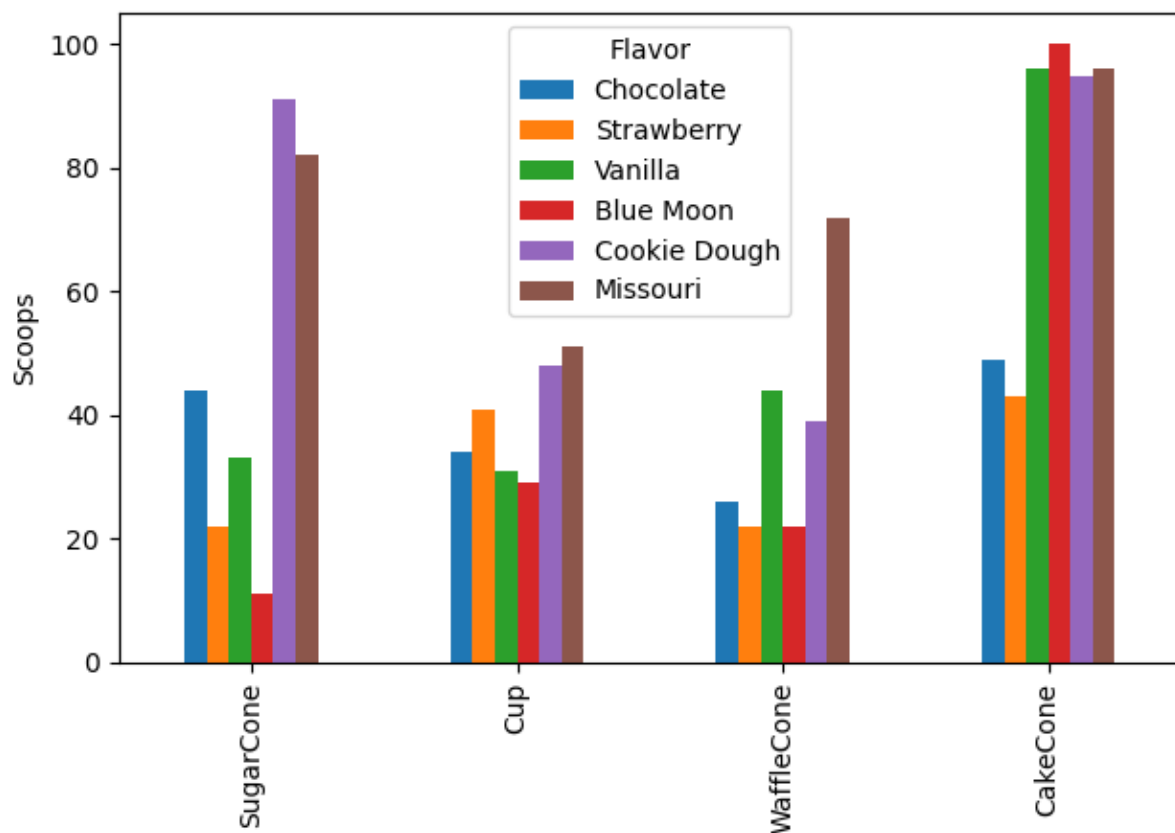
Python 3.10.11 (/Applications/Thonny.app/Contents/Frameworks/Python.framework/Versions/3.10/bin/python3.10)

>>>

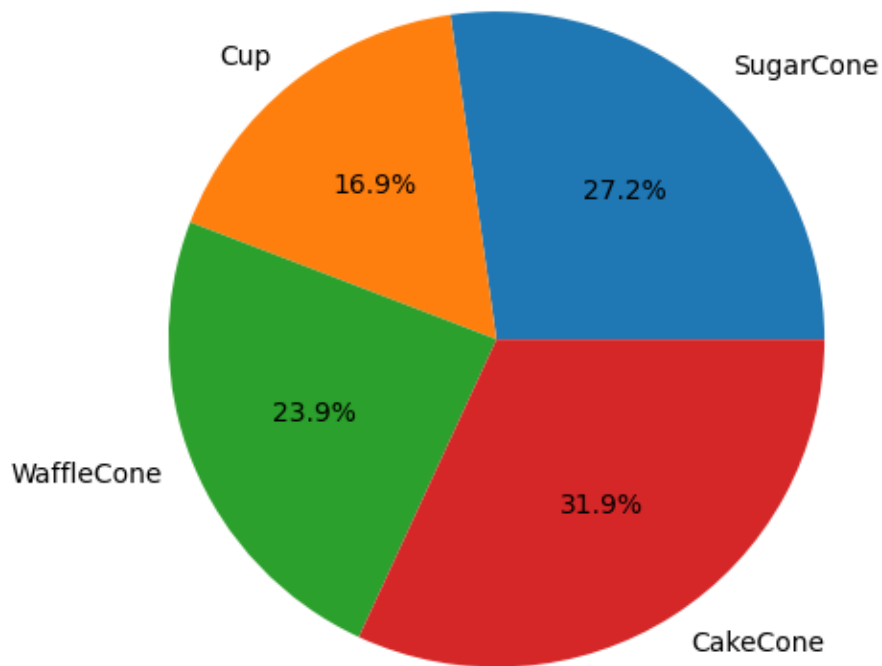
All Flavors and All Containers



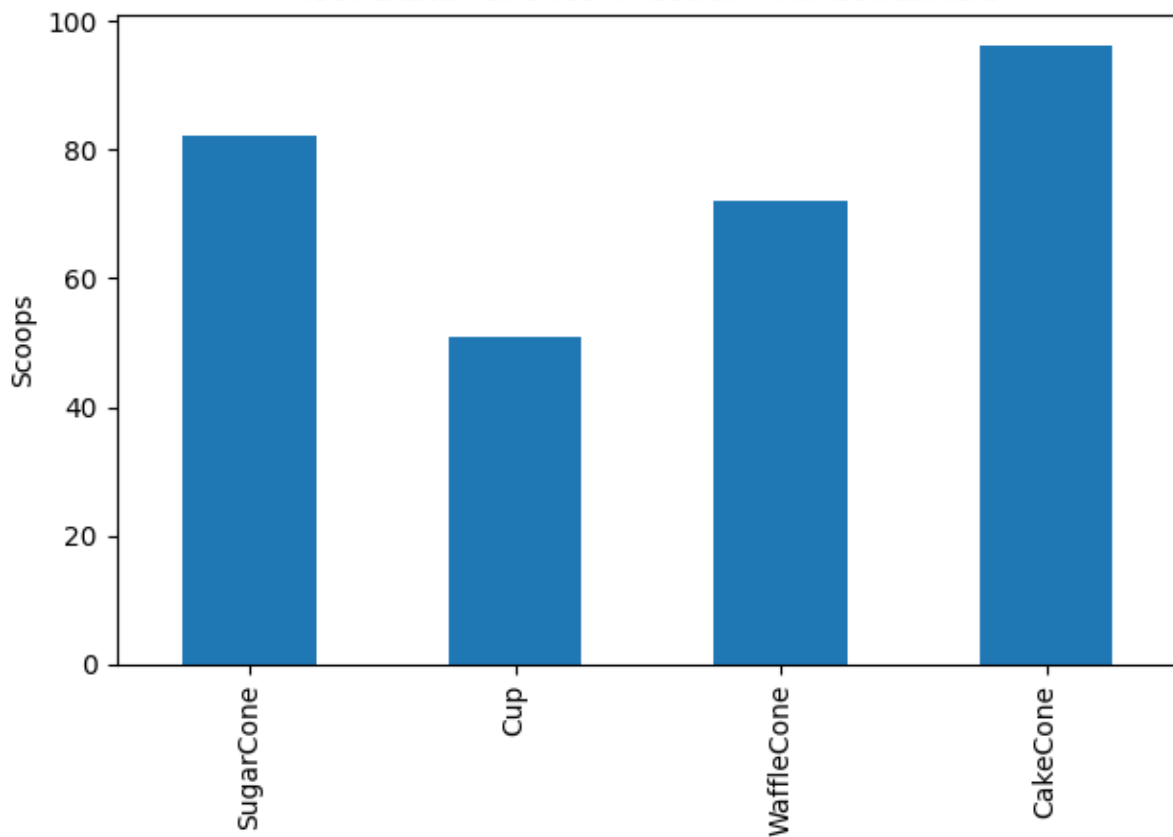
All Containers and All Flavors



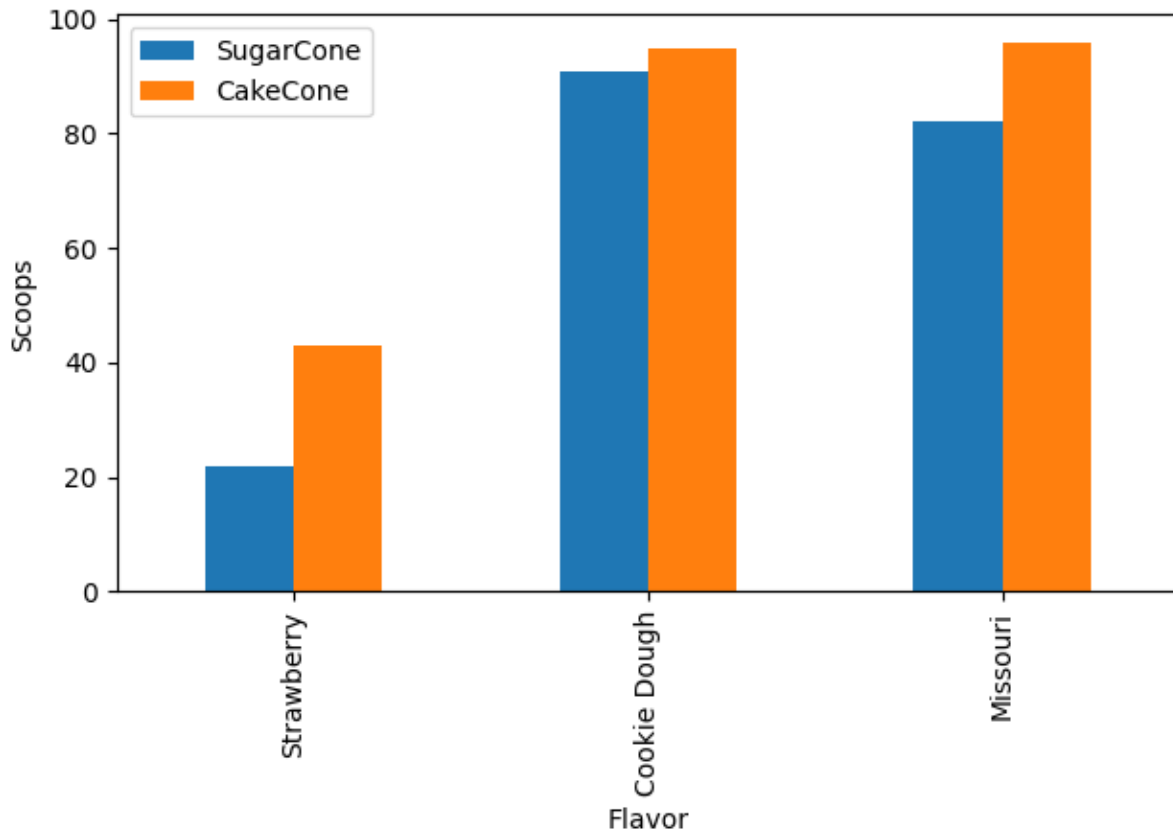
### Your State Favorite: Missouri



### Your State Favorite: Missouri - All Containers



Selected Flavors and Containers



Selected Containers and Flavors

